

ANX-PR/CL/001-01

LEARNING GUIDE

SUBJECT

53002071 - Modelling An Simulation Of Thermal Systems

DEGREE PROGRAMME

05BK - Máster Universitario En Ingeniería De La Energía

ACADEMIC YEAR & SEMESTER

2025/26 - Semester 2

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1. Description

1.1. Subject details

Name of the subject	53002071 - Modelling An Simulation Of Thermal Systems
No of credits	4.5 ECTS
Type	Optional/elective
Academic year of the programme	First year
Semester of tuition	Semester 2
Tuition period	February-June
Tuition languages	English
Degree programme	05BK - Máster Universitario en Ingeniería de la Energía
Centre	05 - E.T.S. De Ingenieros Industriales
Academic year	2025-26

2. Faculty

2.1. Faculty members with subject teaching role

Name and surname	Office/Room	Email	Tutoring hours *
Elisa Alonso Romero		elisa.alonso@upm.es	Sin horario. Request an appointment by email
Javier Rodriguez Martin		javier.rodriguez.martin@upm.es	Sin horario. Request an appointment by email

Javier Muñoz Anton		javier.munoz.anton@upm.es	Sin horario. Request an appointment by email
Susana Sanchez Orgaz		susana.sanchez.orgaz@upm .es	Sin horario. Request an appointment by email
Carlos Rafael Arnaiz Del Pozo (Subject coordinator)		cr.arnaiz@upm.es	Sin horario.

* The tutoring schedule is indicative and subject to possible changes. Please check tutoring times with the faculty member in charge.

3. Prior knowledge recommended to take the subject

3.1. Recommended (passed) subjects

The subject - recommended (passed), are not defined.

3.2. Other recommended learning outcomes

- Álgebra
- Termodinámica
- Transferencia de Calor
- Cálculo

4. Skills and learning outcomes *

4.1. Skills to be learned

CB7 - Que los estudiantes sepan aplicar los conocimientos adquiridos y su capacidad de resolución de problemas en entornos nuevos o poco conocidos dentro de contextos más amplios (o multidisciplinares) relacionados con su área de estudio

CB9 - Que los estudiantes sepan comunicar sus conclusiones y los conocimientos y razones últimas que las sustentan a públicos especializados y no especializados de un modo claro y sin ambigüedades

CE1 - Ser capaz de aplicar conocimientos y capacidades a estudiar, analizar y auditar programas de optimización energética en los diferentes sectores industriales, residenciales, domésticos, plantas de potencia y a la industria térmica y de fluidos en general, en los ámbitos de la eficiencia, la diversificación y la reducción de su impacto en el medio ambiente.

CE11 - Analizar el comportamiento energético y control de los sistemas de energías renovables determinando y aplicando criterios innovadores de optimización energética, económica y ambiental, aplicando metodologías de diseño, simulación y análisis de los componentes y sistemas de energías renovables: solares, eólicos, hidráulicos, de biomasa, de energías marinas, geotérmicas y otras energías renovables; para contribuir a su desarrollo tecnológico y a su competitividad con otras tecnologías energéticas.

CE3 - Utilizar las herramientas necesarias para el diseño y análisis de sistemas de generación, transformación, almacenamiento y utilización de energías nucleares, mecánicas, eléctricas, térmicas e hidráulicas.

CG1 - Aplicar conocimientos de ciencias y tecnologías avanzadas a la práctica profesional o investigadora de la Ingeniería Energética.

CG2 - Poseer capacidad para diseñar, desarrollar, implementar, gestionar y mejorar productos, sistemas y procesos en los distintos ámbitos energéticos, usando técnicas analíticas, computacionales o experimentales avanzadas.

CG8 - Incorporar nuevas tecnologías y herramientas avanzadas de la Ingeniería Energética en sus actividades profesionales o investigadoras.

CT1 - Aplica. Habilidad para aplicar conocimientos científicos, matemáticos y tecnológicos en sistemas relacionados con la práctica de la ingeniería.

CT11 - Usa herramientas. Habilidad para usar las técnicas, destrezas y herramientas ingenieriles modernas necesarias para la práctica de la ingeniería.

CT12 - Es bilingüe. Capacidad de trabajar en un entorno bilingüe (inglés/español).

CT14 - Idea. Creatividad.

CT3 - Diseña. Habilidad para diseñar un sistema, componente o proceso que alcance los requisitos deseados teniendo en cuenta restricciones realistas tales como las económicas, medioambientales, sociales, políticas, éticas, de salud y seguridad, de fabricación y de sostenibilidad.

CT5 - Resuelve. Habilidad para identificar, formular y resolver problemas de ingeniería.

4.2. Learning outcomes

RA53 - Modelizar y simular un sistema energético complejos

RA52 - Modelizar y simular los procesos elementales que intervienen un sistema energético

RA54 - Analizar y proponer mejoras a un sistemas energético existente

RA55 - Utilizar diferentes herramientas de simulación de sistemas energéticos

* The Learning Guides should reflect the Skills and Learning Outcomes in the same way as indicated in the Degree Verification Memory. For this reason, they have not been translated into English and appear in Spanish.

5. Brief description of the subject and syllabus

5.1. Brief description of the subject

Throughout the course of Modelling and Simulation of Thermal Systems, the student will develop the skills to:

- Approach the design of thermal systems through modelling from first principles.
- Identify the degrees of freedom of a mathematical model of a thermal system and introduce specifications for the solving of the model in a simulation tool.
- Incorporate technology specifications and restrictions to the system components.
- Develop a cost estimate and cash flow analysis of the modelled thermal system with a preliminary sizing of its main components.
- Become acquainted with relevant industrial thermal systems and develop skills to represent their behaviour under different operating conditions and a criterion to assess their performance.
- Perform optimization studies of energy systems. Evaluate the deployment of different technologies based on predefined scenarios, load profiles and renewable energy availability data.

The core objective of the subject is to teach engineers not only the basic use of a stationary modelling software but the methods and techniques to approach the design of a wide scope of thermal systems based on mathematical modelling and techno-economic assessment. Also, the students will learn to assess the technoeconomic and environmental performance of the thermal systems designed in an energy system-scale model.

The subject focuses on the synthesis and integration of basic building blocks (unit operations) from a fundamental and technology perspective, subsequently developing a cost assessment to introduce a value function by means of which the design of the thermal system can be evaluated. Therefore, three design pillars are highlighted throughout the course: Fundamentals, Technology & Economics.

3 hours of the subject will be taught per week in one session.

5.2. Syllabus

1. Introduction

- 1.1. 1.1 System Model Simulation
- 1.2. 1.2 Mathematical Modelling of Physical Systems
- 1.3. 1.3 Open systems
- 1.4. 1.4 Simulation tool

2. Thermal Systems

- 2.1. Ideal Carnot Machine
- 2.2. Direct Thermodynamic Machines
 - 2.2.1. Internal Heat Source
 - 2.2.2. External Heat Source
- 2.3. Inverse Thermodynamic Machines
 - 2.3.1. Refrigerators
 - 2.3.2. Heat Pumps
 - 2.3.3. Liquefiers
- 2.4. Chemical Systems

3. Economic Model

- 3.1. Capital and operational cost estimations
- 3.2. Value functions (LCOP, IRR)
- 3.3. Cash flow analysis tool

4. Optimization of Energy Systems

- 4.1. Introduction to linear optimization: Cost function & restrictions
- 4.2. Energy system model in GAMS
- 4.3. Scenario definition, model input & results.

5. Individual Project

6. Applied examples of thermal systems (Case Studies)

6. Schedule

6.1. Subject schedule*

Week	Type 1 activities	Type 2 activities	Distant / On-line	Assessment activities
1	Theory and applied examples of thermal systems Duration: 03:00 Lecture			Progressive Evaluation Activities Problem-solving test Progressive assessment Presential Duration: 00:00
2	Theory and applied examples of thermal systems Duration: 03:00 Lecture			
3	Theory and applied examples of thermal systems Duration: 03:00 Lecture			
4	Theory and applied examples of thermal systems Duration: 03:00 Lecture			
5	Theory and applied examples of thermal systems Duration: 03:00 Lecture			
6	Theory and applied examples of thermal systems Duration: 03:00 Lecture			
7	Theory and applied examples of thermal systems Duration: 03:00 Lecture			
8	Theory and applied examples of thermal systems Duration: 03:00 Lecture	Individual Project Assignment Duration: 00:00 Additional activities		
9	Theory and applied examples of thermal systems Duration: 03:00 Lecture			
10	Theory and applied examples of thermal systems Duration: 03:00 Lecture			

11	Theory and applied examples of thermal systems Duration: 03:00 Lecture			
12	Theory and applied examples of thermal systems Duration: 03:00 Lecture			
13	Theory and applied examples of thermal systems Duration: 03:00 Lecture			
14				Final Presentation of the Individual Project. Model, report and economic evaluation derivables deadline. Problem-solving test Progressive assessment Presential Duration: 03:00
15				Final Exam: Modelling of a thermal system and economic evaluation. Problem-solving test Global examination Not Presential Duration: 03:00
16				
17				

Depending on the programme study plan, total values will be calculated according to the ECTS credit unit as 26/27 hours of student face-to-face contact and independent study time.

7. Activities and assessment criteria

7.1. Assessment activities

7.1.1. Assessment

Week	Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
1	Progressive Evaluation Activities	Problem-solving test	Face-to-face	00:00	50%	4 / 10	CE1 CG1 CB7 CG2 CG8 CT11 CT12 CT3 CE3 CB9 CT1 CT5 CT14 CE11
14	Final Presentation of the Individual Project. Model, report and economic evaluation derivables deadline.	Problem-solving test	Face-to-face	03:00	50%	5 / 10	CT11 CT12 CE1 CG1 CB7 CG2 CG8 CT3 CE3 CB9 CT1 CT5 CT14 CE11

7.1.2. Global examination

Week	Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
	Final Exam: Modelling of a thermal	Problem-					CT11 CT12 CE1 CG1 CB7 CG2 CG8

15	system and economic evaluation.	solving test	No Presential	03:00	100%	5 / 10	CT3 CE3 CB9 CT1 CT5 CT14 CE11
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7.1.3. Referred (re-sit) examination

No se ha definido la evaluación extraordinaria.

7.2. Assessment criteria

The subject will be evaluated in the following way:

PROGRESSIVE EVALUATION

50% Progressive evaluation activities carried out in the classroom, which cannot be recovered and their date may not be programmed beforehand. A minimum of three (3) will be done.

50% Individual project of which:

20% Model in Unisim and Economic assessment spreadsheet

15% Report

15% Final presentation

To pass the progressive evaluation, the following is required:

Given the practical nature of the subject, only three non-justified absences will be allowed.

The minimum grade required in the individual project will be 5/10

EVALUATION ONLY WITH A FINAL EXAM OR EXTRAORDINARY CALL

In case of failing the progressive evaluation, a final exam similar to one or several Case Studies will be carried out. The minimum grade required to pass the exam is 5/10.

8. Teaching resources

8.1. Teaching resources for the subject

Name	Type	Notes
Dhar, P. L. (2016). Thermal System Design and Simulation. Academic Press.	Bibliography	
Dincer, I., Rosen, M. A., & Ahmadi, P. (2017). Optimization of energy systems. New York: Wiley.	Bibliography	
Jaluria, Y. (2019). Design and Optimization of Thermal Systems: with MATLAB Applications. CRC press.	Bibliography	
Ting, D. S. (2021). Engineering Design and Optimization of Thermofluid Systems. John Wiley & Sons.	Bibliography	
Carlos Arnaiz del Pozo, Schalk Cloete, and Ángel Jiménez Álvaro. Standard Economic Assessment (SEA) Tool. Available from: https://bit.ly/3hyF1TT	Web resource	Economic assessments will be carried out with the following Excel tool:

9. Other information

9.1. Other information about the subject

The subject is related to ODS7